

CHAMOUCOUANE, QC, Canada

WMO: 715230

Lat: 49.278N

Lon: 73.356W

Elev: 304

StdP: 97.73

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.3	(c) -34.2	(d) -40.8	(e) 0.1	(f) -37.2	(g) -37.4	(h) 0.1	(i) -34.2	(j) 4.1	(k) -9.5	(l) 3.5	(m) -11.0	(n) 0.5	(o) 10	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.0	(c) 28.8	(d) 19.7	(e) 26.7	(f) 18.6	(g) 24.8	(h) 17.9	(i) 21.7	(j) 25.3	(k) 20.5	(l) 24.4	(m) 19.4	(n) 23.1	(o) 2.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.5	15.8	23.2	19.3	14.6	22.0	18.1	13.5	20.7	64.9	25.5	60.5	24.4	56.7	23.2	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.1	3.4	3.1	-41.6	32.5	2.9	1.7	-43.7	33.7	-45.4	34.8	-47.0	35.7	-49.2	37.0
			-41.5	23.6	2.9	1.3	-43.6	24.5	-45.3	25.2	-47.0	26.0	-49.1	26.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-18.1	-16.6	-9.3	-0.3	8.2	14.2	16.7	15.5	10.8	4.1	-3.7	-12.6
	DBStd	13.45	8.30	7.80	7.97	5.32	4.72	4.35	3.12	3.30	4.33	4.20	5.57	7.38
	HDD10.0	3972	870	746	599	308	93	11	0	2	41	191	412	700
	HDD18.3	6438	1128	979	857	558	315	137	71	100	229	443	662	958
	CDD10.0	624	0	0	0	0	38	137	207	171	65	7	0	0
	CDD18.3	48	0	0	0	0	1	13	19	11	3	0	0	0
	CDH23.3	745	0	0	0	0	62	246	252	149	36	1	0	0
	CDH26.7	182	0	0	0	0	13	74	62	27	7	0	0	0
(14) Wind	WSAvg	1.1	1.0	1.1	1.3	1.3	1.3	1.1	1.0	1.0	0.9	1.2	1.1	1.1
(15) Precipitation	PrecAvg	988	62	52	55	70	74	97	117	100	113	93	79	76
	PrecMax	1159	125	101	107	115	129	149	192	156	206	141	127	171
	PrecMin	826	24	25	28	29	23	48	53	50	63	33	34	39
	PrecStd	81	23	18	20	20	24	26	31	26	32	27	22	28
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.8	3.9	11.7	17.2	28.3	31.9	31.2	29.2	27.4	20.1	10.6	2.7
		MCWB	2.7	2.2	5.7	8.7	16.5	20.7	20.7	20.4	20.9	15.3	9.1	2.2
	2%	DB	0.4	0.6	7.3	14.1	24.8	28.7	28.3	27.0	23.4	16.2	7.7	0.8
		MCWB	0.3	-0.3	3.2	7.2	14.7	19.0	19.8	19.9	18.9	12.9	6.7	0.7
	5%	DB	-2.2	-1.4	4.3	10.9	21.4	26.2	26.2	24.8	20.8	13.0	4.9	-0.3
		MCWB	-2.5	-2.2	1.3	5.4	13.4	17.4	18.8	18.9	17.3	10.9	4.2	-0.5
	10%	DB	-5.0	-3.6	1.9	8.0	18.1	23.5	24.0	22.7	18.4	10.6	2.8	-1.9
		MCWB	-5.4	-4.3	0.1	3.8	11.8	16.3	18.1	17.8	15.7	9.1	2.3	-2.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.6	6.7	10.2	18.6	22.8	23.1	22.7	22.0	16.4	9.8	2.1
		MCDB	2.4	3.2	10.1	14.9	25.1	28.3	27.3	26.2	26.2	19.6	10.2	2.4
	2%	WB	0.2	-0.2	3.7	7.8	16.2	20.6	21.7	21.1	19.9	13.8	6.9	0.6
		MCDB	0.2	0.5	6.6	12.9	21.6	26.0	24.6	24.8	22.3	15.8	7.5	0.7
	5%	WB	-2.8	-2.2	1.8	6.1	14.3	19.2	20.4	20.0	18.0	11.3	4.2	-0.7
		MCDB	-2.6	-1.5	3.8	10.2	19.9	23.5	24.1	23.6	20.2	12.5	4.8	-0.6
	10%	WB	-5.6	-4.3	0.2	4.4	12.6	17.6	19.3	18.7	16.0	9.3	2.4	-2.5
		MCDB	-5.2	-3.7	1.8	7.4	17.5	21.8	23.1	21.6	17.8	10.4	2.9	-2.2
(35) Mean Daily Temperature Range	MDBR		12.7	15.5	15.4	12.8	14.1	14.4	13.0	12.5	11.7	8.4	7.3	9.5
	5% DB	MCDBR	11.3	11.8	16.8	17.6	20.7	18.9	16.9	16.0	15.4	13.7	9.4	7.5
		MCWBR	11.2	10.7	12.9	11.2	11.3	9.5	9.0	9.3	10.2	10.0	8.1	7.7
	5% WB	MCDBR	10.9	10.7	14.2	15.4	17.8	15.3	13.4	13.5	14.2	12.4	8.8	7.7
		MCWBR	10.8	9.8	11.0	10.4	10.3	8.8	8.1	8.6	10.1	9.6	7.9	7.6
(40) Clear-Sky Solar Irradiance	taub		0.265	0.281	0.290	0.342	0.344	0.369	0.386	0.368	0.356	0.320	0.320	0.263
	taud		2.237	2.212	2.247	2.312	2.447	2.405	2.379	2.431	2.451	2.529	2.323	2.291
	Ebn at Noon		787	863	922	904	915	890	868	870	844	824	709	740
	Edh at Noon		73	96	111	117	107	113	114	104	92	72	69	61
(44) All-Sky Solar Radiation	RadAvg		1.30	2.36	3.69	4.68	5.26	5.60	5.12	4.27	3.23	1.85	0.97	0.81
	RadStd		0.16	0.29	0.36	0.38	0.48	0.42	0.47	0.28	0.28	0.18	0.12	0.11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page